



UNIVERSITY COLLEGE OF ENGINEERING SCIENCE AND TECHNOLOGY HYDERABAD
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
Kukatpally, Hyderabad - 500 085. Telangana State, INDIA
MID EXAMINATION ANSWER BOOKLET

H.T. No.

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NAME OF THE EXAMINATION Mid-Term Examinations.SUBJECT Data-Base Management Systems [DBMS]BRANCH/SPECIALIZATION CSE-SE [IDDMP] [2nd Year 2nd Semester]DATE OF EXAM: 06/03/2025.

Signature of Invigilator

Please tick (✓) the questions and bits answered by you
in the space provided above the dotted line

	1	2	3	4	5	6	7	8	9	10
Part A	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒

Part B

Question No.	1	2	3	4	5	6	7	8
Tick Marks	☒	☒	☒		☒			
Tick Marks								
Tick Marks								
Question Wise Total Marks								
(Part A + Part B) = Grand Total (in Figures)								
in words								

INSTRUCTIONS TO CANDIDATES OVERLEAF

Q1) The File System:

The File System is a place where the information is stored in the Files.

The Data - Base Management System:

The Data-base is where the information is stored in the Data-base.

	File System	DBMS.
Structure	The Structure of the File System says that the information or the data is stored in the form of Files.	The Structure of DBMS states that the information or the data is stored in the Data base.
Data.	The Data manipulation is Possible.	The Data manipulation can't be Possible.
Backup.	No - backup data is Possible in File System.	Data backup is Possible in Data-base Management System.
Query	No - Query's can be Solved in the File System.	Query's can be Solved in the DBMS.
Consistency.	Data Consistency is not Possible. As multiple File will be created.	The DBMS will have a consistent Data.
Complexity.	No - complexity as creating a File and storing data in it is easy.	storing data in DBMS is complex.

	File System	DBMS.
Security.	There will be less Security and the information can be easily modified	The Security in DBMS is very tight.
Cost	The File System is very less expensive.	The DBMS is very Expensive.
Data Independence.	The Data is not independent.	The Data is independent from other higher levels
updation.	The To add data we need to create multiple Files.	The Data can be easily updated.
Sharing	The data can't be shared as multiple File are Present	The data can be shared easily.
user access	An one can access the data from the File System.	A The Data-Base Administrator should give Permission to access the Data.
Data Intendency (or) Rules.	There are no rules.	There are rules.
Examples.	Notepad, word etc---	MySQL, Oracle.

- From this we can say that the Data-Base Management System is far more better than traditional File Systems.

- As the File system has multiple Draw-backs compared to DBMS.

- The DBMS is very useful and upgraded version compared to the traditional File System.

Q2) Fundamental operations of Relational algebra:

- The Relational algebra is used to find or indicate the relation between two attributes of an Fundamental operating system.

- Selection (σ)
- Projection (π)
- Union ($\cup$)
- difference ($-$)
- Cartesian Product ($\times$)
- Rename (ρ)

* Selection (σ): The Selection function is used to select the row from the table.

Symbol: σ

In Selection we use from, where.

⇒ Example: Select * from emp;
where age > 20;

* Projection (π): The term Projection is also the Relational algebra. It is used to select the columns (or) entities from the table.

Symbol: π

Example: Select from empl,
where (π -name, age);

Union: The Union function is used to combine the two same entities from two different tables.

Symbol: $\cup$

Example: Select from empl, dept
where (σ dept-no from empl) $\cup$ (σ dept-no from dept)

Difference: The Difference function is used to find the different entities from two different tables.

Symbol: $-$

Example: we can use this to find the different from

Select from student-table,
where (Students - Student-Graduated);

Cartesian Product: The cartesian Product is used to Product the entities from two different tables.

Symbol: $\times$

Rename: The Rename function is used to rename the entity from a table.

Symbol: ρ

Example: Select from emp;
where (student_name = p(student_id));

3. Design E-R model for a banking enterprise.

E-R model: The Entity-Relational model is where the real-time problems are shown in pictorial form.

The E-R model consists of 3 parts:

Entity, Attributes, Relationship.

Entity: The column in the table are known as Entities. The entities should be a real-time objects.

Ex: Place, Teacher etc...

Attributes: The rows column in the table is known as Attributes (or) tuples. The Attributes will tell the word of the entities.

Ex: Teacher teaches. etc..

Relationship: The relationship states the relation between the Entity and Attributes.

E-R-Diagram of banking Enterprise:

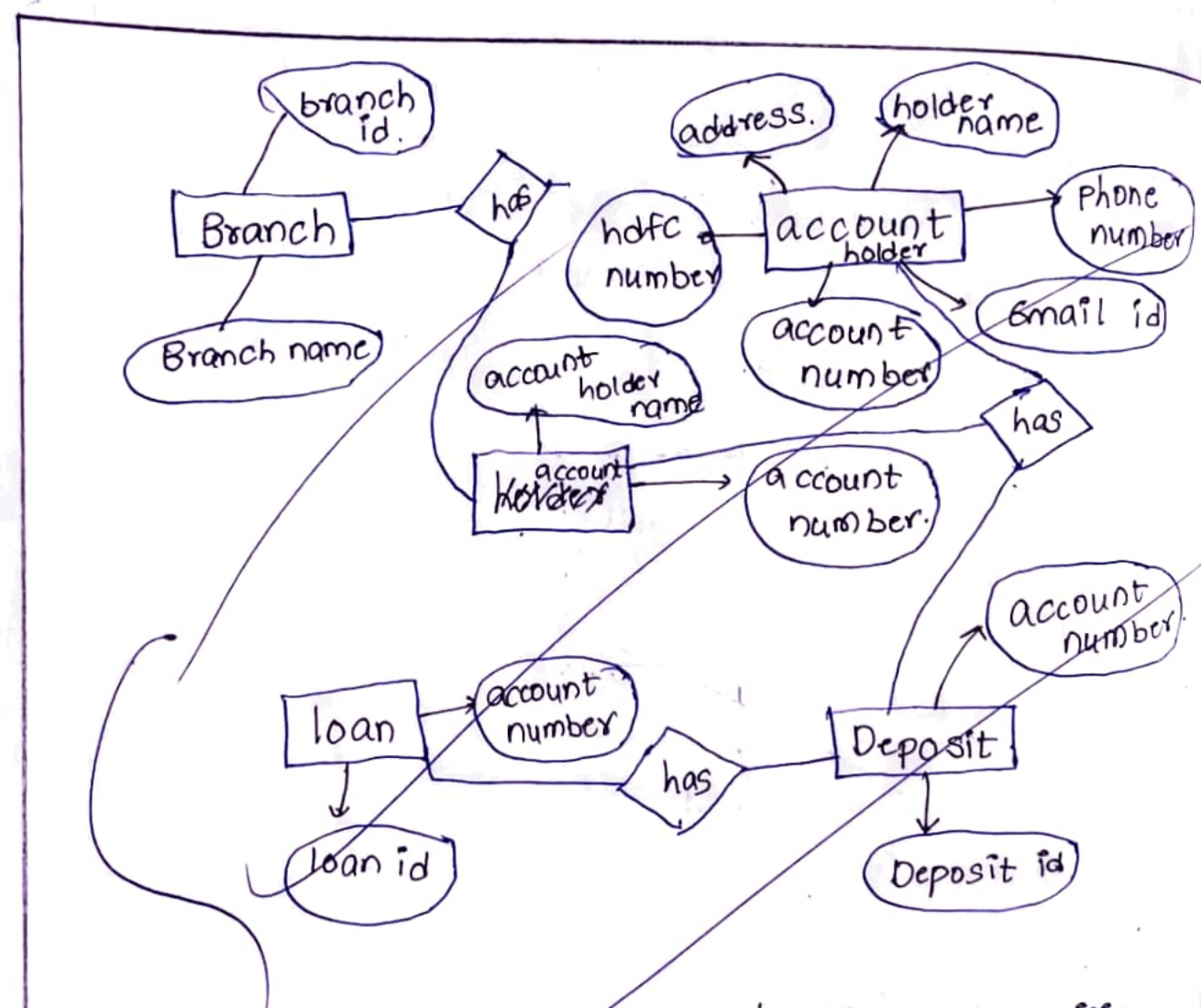
In banking enterprise the banks has the entities such as accounts, holders, branches, loan, Deposits.

In E-R-diagram the entities are represented using "Rectangular shape".

The attributes for the Entities are account-id, holder-name, phone-number, Email, hdfc code, branch name, deposit-number, deposit-id; loan id.

The attributes are represented using sphere shape.

The Relationship of the attributes and Entities is represented using diamond shape.



The ER Diagram of banking Enterprise.

The relationships are:

The Branch has accounts.

The account has account holders

The account holder has Deposits and loans.

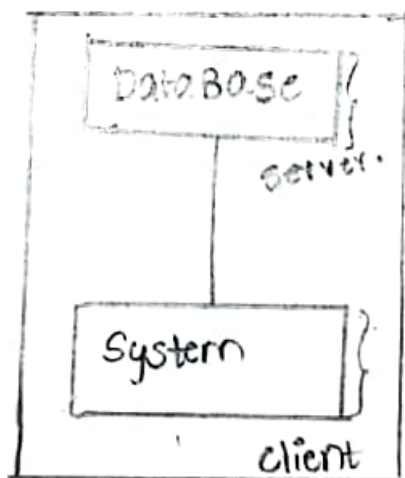
Q5) The Data Base - Architecture:

The Data - Base Architecture represents the Schema of the Information or the Data.

The Data base- Architecture are of 3-types:

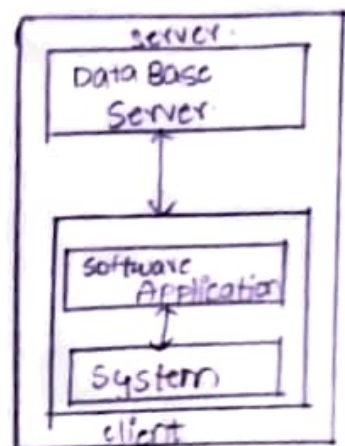
- 1-tier Architecture
- 2-tier Architecture
- 3-tier Architecture.

1-tier Architecture: The 1-tier Architecture is where the System and the Database are directly connected to each-other. The system access the Data from the data Base directly.



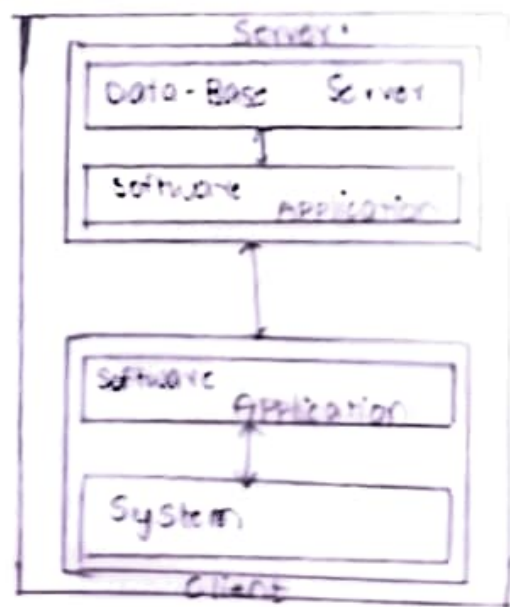
2-tier Architecture: The 2-tier Architecture is where the System doesn't directly contact the DataBase. It has an Application Software in the client side and the Application Software contacts with the Server DataBase.

Example: The Railway Reservation follows 2-tier

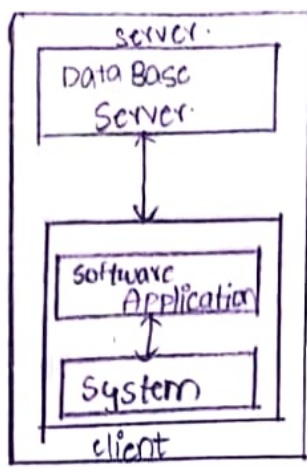


3-tier Architecture: The 3-tier Architecture is where the Data Base Server will also have an Software Application.

The System will interact with Software Application of System and the System Software Application will interact with the Software application of Data Base and which interacts with the Data Base.

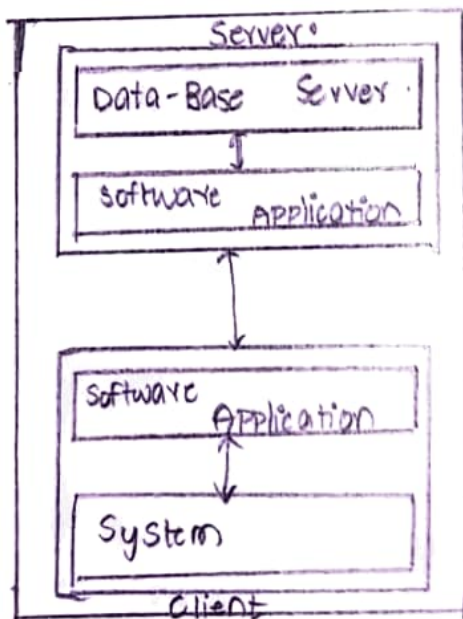


Example: The Amazon, Flipcart etc.. use this type of Architecture.



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